

Rahul Chauhan

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Summary

Curious, Result-Oriented, Techie aspiring to craft an impactful career in a Modern, Data-Driven Software Organization with focus on AI-Driven Solutions and Advanced Analytics. 2+ years of Industry and Research Experience using Python, R, Excel, BI, and Tableau.

Work Experience

99P labs, Honda Research Institute, CA, USA 01/24 – 05/24
Data Scientist

- Architected a data pipeline for a telematics dataset of over 1M+ rows, developing driver profiling and behavior scoring systems with Isolation Forest, XGBoost, and ANN, yielding a prediction accuracy of 72.39%.
- Designed a Power BI Dashboard for vehicle mobility management, visualizing fuel efficiency, cabin comfort, and speed management to aid decision-making by 20%.

Leeds School of Business, University of Colorado Boulder, CO, USA 01/24 – 05/24
Graduate Research Assistant

- Utilized NLP Techniques (Word2Vec, VADER, LDA) to analyze 500+ comments on corporate gun violence stances, achieving 80.67% prediction accuracy.
- Automated the collection and literature review from 1500 market research papers using document parsing algorithms, reducing manual effort by 90% and accelerating research process by 35%.

LTIMindtree (Formerly known as Larsen & Toubro Infotech), MH, IN 06/21 – 07/22
Functional Consultant

- Leveraged SAP's Plant Maintenance to conduct data-driven analysis, identifying trends and predicting maintenance needs, which reduced equipment downtime and increased production efficiency by 12%.
- Led the OneSAP system integration for Legacy Systems, using data analytics to streamline project management process, assess risks, and enhance decision-making, resulting in successful transformation with a 40,000 euros turnover increase.

Illustrative Projects

Multivariate Time Series Analysis for Temperature Forecasting — Python [Link]

- Engineered a 64-unit LSTM-based neural network, 7-time steps and 6 features in TensorFlow for temperature forecasting based on historical data collected over 5+ years. Achieved a training RMSE of 0.0279 and validation RMSE of 0.0296.

Harnessing AI and Prompt Engineering for Intelligent Chatbot Development — Python [Link]

- Innovated a Python-based chatbot using LangChain & GPT API to automate bank appointment scheduling, integrating Zapier for 50k+ app actions, reducing scheduling time by 70%.

Insights on Delta Airlines using Machine Learning — R [Link]

- Created a data pipeline and implemented multiple machine learning models to analyze 20,000+ Delta Airlines reviews. Achieved 71.56% prediction accuracy for customer feedback, identifying a 20% recommendation rate with an average rating of 3/5.

OCD Patient Data Analytics and Visualization — SQL [Link]

- Built a Health Analytics Dashboard with SQL and Excel for analyzing OCD data of 1,300+ patients, enhancing decisions with gender and ethnicity insights. Streamlined monthly reporting and reduced manual data handling by 50% using SQL & Power BI.

Credit Card Transactions Grievance Dashboard — Tableau

[Link]

- Developed a Tableau dashboard for real-time tracking of 86,893 credit card fraud complaints, utilizing data visualization, weekly trends, and geographical representation to enhance response times and strategic decision-making.

Education

University of Colorado Boulder, CO, USA <i>Master's in Data Science</i>	<i>08/22 – 05/24</i> GPA: 3.83/4.00
University of Mumbai, MH, IND <i>Bachelor's in Information Technology</i>	<i>08/17 – 05/21</i> GPA: 8.39/10.00

Publications

- Chauhan, R. (2024). Yay, it's Honda Telematics Data Explorer. Medium. [Publication]
- Chauhan, R. (2021). Online food ordering and delivery system: Connecting local vendors to customers. SSRN. [Publication]

Trainings and Certifications

Building Multimodal Search and RAG, Prompt Engineering for Vision Models, Preprocessing Unstructured Data for LLM Applications, JavaScript RAG Web Apps with LlamaIndex, Serverless LLM apps with Amazon Bedrock, Automated Testing for LLMOps, Microsoft: Introduction to C#, SAS Programming, LangChain: Chat with Your Data, Diffusion Models.

Skills

Programming & Databases: Python, R, C#, JavaScript, SAS, SQL (MySQL, PostgreSQL, Cassandra DataStax)
Data Science & Machine Learning: Statistical Analysis, Machine Learning, Neural Networks (LSTM, GRU, CNN, ANN), NLP (NLTK, VADER, Word2Vec), Predictive Modeling, Time Series Analysis, MLOps (CI/CD, model monitoring)